involves producing fewer consumer goods and freeing resources for the production of investment goods. This can also be represented diagrammatically as in Fig. 2.7, where equilibrium income $Y_e$ is attained at the intersection of the saving function and the investment function.

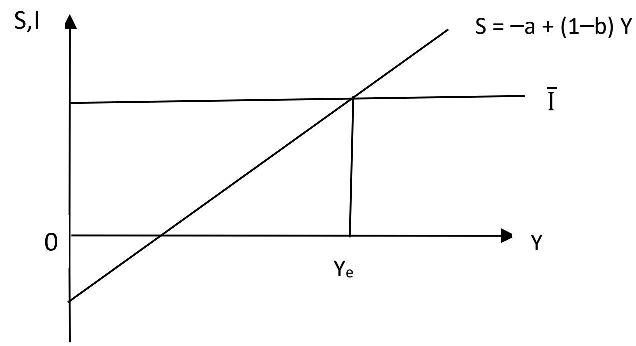


**Fig. 2.7: Saving-Investment Equality**

In a three-sector economy, the output market equilibrium condition is $Y = C + \bar{I} + G$. Income has three uses: consumption, saving and payment of taxes. Thus, $Y = C + S + T$. Putting these together, we can write the output market equilibrium condition as:

$$C + S + T = Y = C + \bar{I} + G, \text{ or, } S + T = \bar{I} + G \qquad \dots (2.14)$$

This can also be written as $S + (T - G) = \bar{I}$, which shows that at equilibrium in a closed economy, planned investment must be equal to total saving that include planned saving by households and public saving $(T - G)$. We can also re-write (2.14) as, $(S - \bar{I}) = (G - T)$. An implication of the above is: when planned investment is less than planned saving $(S - \bar{I} > 0)$, the government is running a budget deficit $(G - T > 0)$. The financing of the budget deficit is by borrowing from households.

### 2.4.3 Paradox of Thrift

An important result regarding the role of saving in the Keynesian model is referred to as the 'saving paradox' or the 'paradox of thrift'. You know that saving on the part of a household is considered as a virtue. Higher saving increases the assets and wealth of a household in the long run. For the economy as a whole, however, an increase in saving may not be good; it may be a vice. When households plan to save more, aggregate saving either remain the same or may even decrease. Let us explain it through a diagram (see Fig. 2.8).

In Fig. 2.8, equilibrium in the economy is at point A, where S = I. Output produced is $Y_0$. Suppose there is an increase in the autonomous component of households' planned saving, i.e., at each level of income planned saving is higher. This involves an upward shift of the saving function from $S_0$ to $S_1$. Note that $S_1$ is parallel to $S_0$ since there is no change in MPS. Now, the equilibrium output level increases to

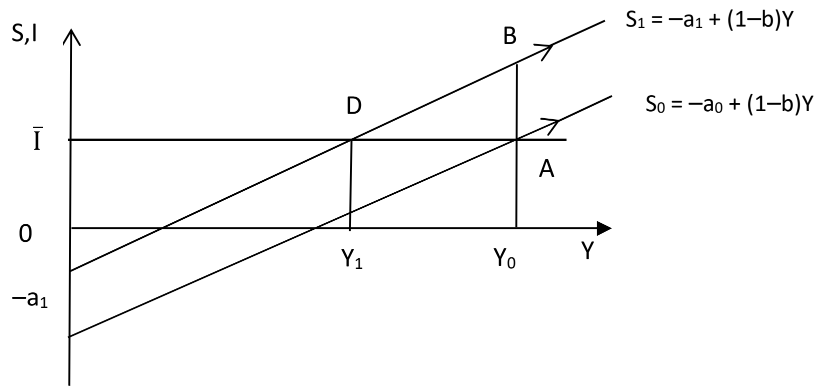


**Fig. 2.8: Paradox of Thrift**

Since there is no change in planned investment, there is disequilibrium in the economy now. Planned investment ($DY_1$) is the *same as before* ($DY_1 = AY_0$). Since people now wish to save more (saving function has shifted to $S_1$), income must be lower, so that saving is the same as before. The paradox is, despite an increase in thriftiness, there is no change in aggregate saving.

The desire to save more reduces planned consumption, thereby lowering aggregate demand and hence output (and income). At lower income levels, saving is correspondingly lower – instead of $BY_0$, aggregate saving fall to $DY_1$, since income has fallen to $OY_1$.

If the investment function has an induced component and is given by $I = c + d\,Y$, as in (2.2b), then an increase in planned saving can actually lead to lower level of saving at equilibrium, as shown in Fig. 2.9 below.

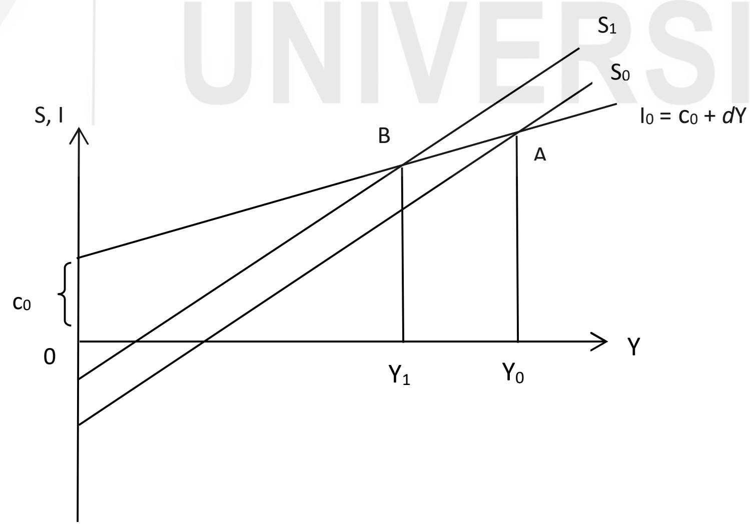


**Fig. 2.9: Paradox of Thrift (Induced Investment)**

In Fig. 2.9 the saving function shifts from $S_0$ to $S_1$. In this case, the reduction in consumption demand (due to rise in planned saving), leads to lower output which further leads to a decrease in induced *s*, thereby lowering income further. At the lower level of equilibrium income ($OY_1 < OY_0$), planned saving is correspondingly lower ($BY_1 < AY_0$), with the desire to save more leading to lower aggregate saving at equilibrium.

This result provides an important insight into the role of saving in the short run. In an economy that faces no supply constraints and has excess production capacity, an increase in the desire to save essentially lowers output by lowering aggregate demand. Therefore, at a lower equilibrium income level aggregate saving is the same or may even be lower than before. Note that such decline in output due to higher saving could be a short run phenomenon. In the long run, higher saving allows for higher investment and hence for creation of additional productive capacity. You will learn more about the role of saving in the long run while studying economic growth.

**Check Your Progress 3**

1. (a) In a two-sector economy (with only households and firms) derive the output market equilibrium condition in terms of planned saving and planned investment.

   (b) In a three-sector economy (with households, firms and government), what would be the relation between saving and investment at equilibrium if the government runs a budget surplus?

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2. What is meant by the 'paradox of thrift'? Use appropriate diagram to explain the cases of (i) investment demand is autonomous; and (ii) investment demand has an induced component.

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